

Organizing IT to promote agility

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Abstract Agility at the individual business unit and organizational levels presents a challenge for many information system (IS) departments. Business unit agility demands the ability to sense and respond to changes in local competitive environments, whereas organizational agility demands the ability to sense broader market opportunities and respond with changes that are organization-wide. The former requires experimentation and customization of system designs, while the latter demands uniformity and standardization. Although flexible and customizable software might assist organizations in adapting systems to meet these challenges simultaneously, coordination across multiple business units also demands effective organization and governance of system design and development. This paper presents a longitudinal case study of an insurance company that was effective at sensing and responding to changes in the environment at the business unit level, but less effective at sensing and responding at the organizational level. Using the platform logic as a theoretical lens, we analyze this case and offer insights into how multi-unit organizations can manage system design at the organizational and business unit levels, thereby supporting agility through the development of effective organizing and governance mechanisms. Our analysis outlines relational and integration mechanisms and explains how these arrangements helped

the organization to attain greater enterprise agility and support its overall strategy.

Keywords Agility · Platform logic · Governance

1 Introduction

Advances in information technology (IT) have allowed businesses to automate the design, sales and support of their products and services. IT also enables enterprise agility or the ability to sense and respond to changes in the competitive environment [1–5]. Reciprocally, an organization's ability to make changes to the design of its products and services becomes dependent on its ability to design and implement changes to the information systems in which those products and services are imbedded. As a result, realizing enterprise agility becomes dependent on how effectively an organization designs and manages its information technologies.

Designing and maintaining IT for enterprise agility is complicated when the competitiveness of the organization's products and services is dependent on an ability to respond to market changes at both the business unit and organizational levels. From a system design perspective, agility at these two levels calls for different approaches. Agility at the business unit level is enhanced when individual units are given autonomy in deciding how to design and implement systems changes in response to the local competitive environment [6]. Alternatively, creating a standardized approach to the design of systems used across business units enables greater agility at an organizational level, since system changes can be implemented en masse.

Managing enterprise agility is considered as a multi-level organizational problem in this paper. Since multi-unit

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organizations are concerned with both local and organization-wide markets, they must manage change at both levels. Complications arise because actions taken at the business unit level affect the organizational level, and vice versa. For example, standardizing all product design at the organizational level would restrict agility at the business unit level, by reducing the ability of business units to respond to local business conditions.

At the business unit level, we consider an organization's ability to sense and quickly respond to local market-specific opportunities central to its ability to achieve business agility. At an organizational level, we consider the same organization's ability to sense and respond to broader market opportunities that may demand wholesale revisions to the design of their products and services, and therefore necessitate organization-wide system changes.

We address this topic by focusing on the following research question: how can multi-unit organizations, where product and service design changes are dependent on information systems, enable agility at both the local business unit and organizational levels? To address this question, we present a longitudinal case study of an insurance company that was very successful in its efforts to achieve faster delivery and customization in the design of its digitized information products and the information systems that supported them at both the organizational and business unit levels.

Our focus in this research is in understanding theoretically how one successful organization was able to develop mechanisms to organize and govern IT in a way that improved its enterprise agility. We present a case study of an organization that struggled to manage both of these challenges simultaneously. Like many IS departments, information technology was at times viewed as a barrier to change in this organization, rather than as an enabler of innovation. The problems the organization struggled with were in part due to its aging and complex systems. More fundamental to their problem, however, was the company's approach to organizing and governing its system design and development activities, which adversely affected its ability to respond at the organizational level when desired.

We do not discount that recent advances in information technologies have helped organizations, including the one in our case study, to become more agile. For example, many organizations have benefited from creating system designs that are mass-customizable [7, 8]. Yet, mobilizing and implementing these designs also requires effective organizational structures in order to manage the delivery processes and maintain system-wide standards over time [9].

The paper begins with theoretical background, examining the various approaches to organizing IT, integrating IT-business processes, and the implications they hold for

developing enterprise agility. The paper then discusses the importance of governing and organizing systems that promote enterprise agility. The platform logic is presented as a theoretical lens for analyzing various mechanisms to develop information systems capabilities [10, 11]. This framework provides insight into how organizations assemble and reassemble combinations of IT capabilities through the use of various organizing mechanisms. As an organizing logic, the framework moves beyond simple structural explanations of governance and instead explores relational and integration mechanisms as arrangements to enable the delivery of capabilities.

Next, the case study is presented, followed by an analysis in which relational and integration mechanisms are identified. These mechanisms provide a theoretical explanation for how the company in the case study was able to respond to demands of developing and delivering product design changes en masse across all 48 of its business units. In the data analysis and discussion, we explain how the platform logic contributes to the IT literature on IT governance and IT agility by connecting the management of enterprise agility to the management of system design and development at both the local and organizational level. Finally, the conclusion summarizes how this framework informs existing research and explains our contribution to current practice by informing managers as to how they can develop more agile information systems design and development processes.

2 Theoretical background

2.1 Governing and organizing product management

Coordination of the IS department has long been a focus of theoretical and practical concern. These concerns are further complicated in many organizations, where the design of products and services need to be tailored to the local competitive market. The need to sense and respond to market conditions requires organizational structures that facilitate effective environmental analysis. These organizations also benefit from efficiently integrating across functional departments to design, develop and deliver the product or service to market [12]. As a result, these firms often organize into many individual business units, servicing different markets with different versions of a product design. The units are designed around product lines or geographic regions. In the product organizational form, individual product managers have control over a number of functional areas, such as pricing, marketing, product design and production.

As business units seek to increase their degree of fit with their own environments, their products, services, processes

and routines must often vary in design and execution [13]. While varying the design of products and services may contribute to the individual business unit's performance, it may come at the expense of organizational efficiency with respect to information systems. Excessive variation can create a source of complexity in systems. As a result, adapting systems to local business unit needs can conflict with the desire to standardize systems, making them easier to alter on a global basis.

Most organizations, even those with an array of individual and autonomous business units, benefit by having shared resources, such as IT, thereby attaining greater economies of scale and scope. However, differences in the way departments design work processes and develop information systems knowledge presents integration challenges for intra-organizational coordination [12]. Since the efforts intended to maximize individual business unit performance may ultimately impact the overall organization's agility, management of IT resources is a central issue of concern. To address this issue, we next discuss how governance mechanisms inform the management of system designs in decentralized multi-unit organizations.

2.2 Governance, structural overlays and organizing information systems

The capability to sense and respond both locally and globally raises questions of how best to govern and organize the design, development and delivery of information system changes in complex organizations. Similar questions are discussed previously in the IT literature in terms of IT governance, organizational structures and horizontal mechanisms [6, 13–15].

Governance of the IT function determines where decision making authority resides in the organization [15]. Governance is traditionally discussed in terms of centralized, decentralized, or federal structures [6, 15, 16]. Governance over decision making holds implications for developing either efficiencies through centralization or greater effectiveness in serving local competitive needs through decentralization. A compromise position, the federal or hybrid structure, allows organizations to centralize shared functions such as infrastructure, while decentralizing others such as application development and its use [15].

Supplementing traditional governance structures are horizontal mechanisms. Such mechanisms are increasingly used to span multiple functions and business units and increase intra-organizational collaboration [14]. Often operating outside the formal communication hierarchy, horizontal mechanisms such as liaisons, steering committees and cross-functional teams create lateral capabilities that facilitate communication and collaboration in such

efforts as information system and product development [14, 17–19].

Governance structures in concert with horizontal mechanisms are often deemed necessary for coordination and prioritization in product organizations [14, 18]. In situations where business unit competitiveness depends on responsive systems, horizontal mechanisms such as liaisons and steering committees can play a critical role in the control of and adherence to standards, influencing resource allocation and project scope.

Additionally, changes in competitive demands can alter IT's role in an organization, highlighting the delays associated with existing variations in design across a single systems that supports different business units. At times, coordination and delivery of innovative designs may necessitate governance mechanisms that can supersede the interests of individual business units in order to insure the achievement of organizational objectives. To date, this challenge presents an under-researched area in the literature regarding IT governance and agility.

2.3 Agility and information systems

Various online dictionaries characterize agility as being nimble, quick and alert. Although an accepted definition has yet to emerge in the management literature [20], agility can be characterized as the ability to sense and respond to changes in an organization's internal and external environment by quickly assembling resources, relationships and capabilities [3, 4, 20]. The underlying components of agility focus on the organization's capabilities to *sense* anticipated and unanticipated changes in the internal and external environment, and to *respond* to those changes [3, 4, 21].

It is the need to sense and respond to these changes that drive organizations, especially those that operate in dynamic, hyper-competitive environments, to develop capabilities that support agility [3, 4, 20, 21]. Changes can originate from the organization's external environment (as a result of social or legal pressures, mergers and consolidations, competitive pressures, customer needs, or emerging technologies [3, 20]) or changes can originate internally (from new systems implementations, shifts in IT governance, and changing IT strategies [20]). Regardless of the source of change, agile organizations are those that are quick to identify and adapt to the impacts of such changes and resourcefully apply existing or create new capabilities, relationships and processes in their responses [22].

IT plays an important role in the agile organization by increasing its ability to sense and respond [3]. An organization's IT capability should be sufficiently advanced so as to sense opportunities that are brought about by changing technologies. For example, changing technologies created

market opportunities for organizations to deliver digitized products, such as music and movies. IT capabilities have enabled many organizations to sense changing market conditions via data mining and analysis techniques [3]. IT capabilities can also increase an organization's capacity to respond to environmental changes and capitalize on opportunities in information-intensive industries, such as insurance and financial services [3, 4]. Moreover, agile IT architectures and infrastructures allow for rapid customization and modification of systems and the products and services they support, thereby providing agile organizations the capacity to explore and exploit market opportunities [2–5, 20].

However, just as IT plays a role in enabling agility, it also plays a role in *disabling* agility [5, 20]. While well-architected systems enhance an organization's ability to access and synthesize information in a timely manner [3, 4], poorly architected systems tie up resources, consuming time and money while limiting responsiveness [20, 22]. Furthermore, investing in the wrong IT infrastructure and systems may limit flexibility and constrain an organization's range of responses [3–5, 20]. For example, many organizations that heavily invested in systems which were designed to provide flexible internal processes were constrained with the emergence of B2B and other inter-organizational e-commerce initiatives [20, 23].

While IT infrastructures and architectures may provide a platform that facilitates an organization's response capability, management actions and policies also play a key role in achieving agility [5]. Given the complexity of simultaneously addressing uncertainty at the organizational and business unit level, combined with changing technologies, organizations often implement standards as a basis from which to increase response capabilities [5]. As in the case presented later, managing standards can represent a mechanism for controlling changes to a system that might otherwise create complexity and slow response capabilities over time [20].

An especially challenging situation exists when managing standards in a multi-unit organization, given that such organizations often require a system to support local-level flexibility, while maintaining a global level platform. Effective organization and governance of IT can assist in this matter by helping organizations to control the implementation of systems designs in a way that facilitates adherence to and adoption of standards. Since changes will almost certainly occur to any set of standards, in light of changing strategies and environmental pressures [3, 20], standards cannot in and of themselves be a permanent solution. The solution must also include effective management of the standards themselves. Organizational structures and governance mechanisms provide a way of understanding management's role in organizing effective

means towards achieving standards and their adaptation, as shifts in business objectives require.

2.4 Platform logic and an organizing framework for agility

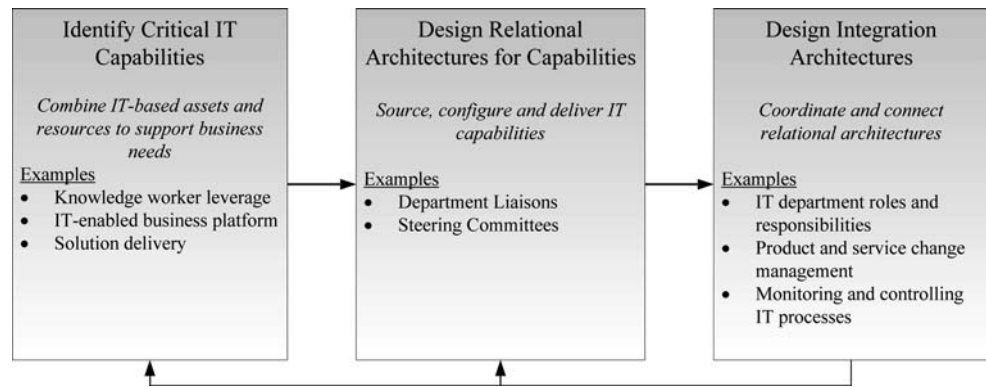
How best to organize the IS function in order to design and adapt information systems to support changes in product and service designs is an important question that is not answered by simple structures, such as governance models and horizontal mechanisms [10, 24]. Similarly, viewing the organization of IT as centralized or decentralized structures has limited the ability of past research to inform theory and practice [10]. These approaches are especially limiting given that organizations increasingly view the development of dynamic capabilities as imperative to their competitive strategy [25]. Thus, choosing the right organizing structure is complicated in a world when organizations are relying on IT to implement strategic change [10].

Some recent research has questioned the salience of governance structures as the sole, or even major, factor in organizing the IT function [10, 11, 24]. As a result, the platform logic is proposed and has been demonstrated to be an alternative approach to theorizing and modeling how IT is organized [10, 11]. The platform logic represents a departure from traditional views of governance. Platforms offer an organizing logic or managerial rationale for creating flexible arrangements of capabilities and relationships that allow organizations to respond to challenges and exploit opportunities [10, 11, 24, 26]. This platform concept explains how organizations can promote IT capabilities by combining and recombining resources through the development of various mechanisms.

Observing that competition and emerging innovations force organizations to be increasingly agile, Ciborra noted that organizations may respond by appropriating and recombining various schemes of routines, resources and structures [26]. This approach allows an organization to remain flexible, tailoring their product and service offerings to exploit ephemeral opportunities [26]. By utilizing the platform logic, organizations achieve agility by developing the capabilities to appropriate resources in response to environmental change. In this context, management's role is to combine and recombine existing resources to meet current demands. In effect, management does not "reinvent the wheel"; they leverage existing structures and resources in novel ways as a basis from which to experiment.

One framework that describes the process for creating platforms for organizing IT activities is comprised of three steps: identifying IT capabilities, designing relational architectures, and designing integration architectures

Fig. 1 Elements of the platform logic for organizing IT activities (adapted from [10])



[10, 11]. The framework and associated examples are presented in Fig. 1. The three steps in the framework are discussed below.

Drawing on the resource-based view of the firm, IT capabilities are “combinations of IT-based assets and routines that support business conduct in value-adding ways” [10]. Viewed another way, a firm’s IT capabilities represent its ability to effectively combine, recombine and deploy the IT resources at its disposal [27]. It is this capacity to leverage existing IT resources that creates competitive advantage [28]. Prior research on IT capabilities has identified the following capabilities: value innovation, knowledge worker leverage, IT-enabled business platform, operational excellence, value-chain extension, solution delivery, and alliance management [10]. Once an organization determines its IT capabilities, it can design the appropriate relational architectures, or mechanisms, two terms we use interchangeably.

Relational mechanisms represent the array of relationships that allow organizations to source, configure and deliver IT capabilities [10, 11]. Examples are steering committees and departmental liaisons [11, 14]. These relational architectures may be either intra- or inter-organizational. Intra-organizational relational architectures are relationship mechanisms within organizations that operate to deliver IT capabilities [11]. Alternatively, inter-organizational relational architectures represent relationship mechanisms that reach beyond conventional firm boundaries to source and configure IT capabilities, such as strategic alliances and joint ventures [10, 11].

Sourcing, configuring and delivering IT capabilities via relational architectures represents a departure from traditional conceptualizations of governing IT capabilities. Rather than viewing governance as having a static locus of decision making (as in the case of centralized, decentralized or federal control), relational architectures account for multiple arrangements for sourcing and configuring IT capabilities [10]. These dynamic “networks of relationships” allow for greater flexibility and agility in acquiring, combining and deploying required resources.

The next step in the framework in Fig. 1 is the design of integration architectures. These mechanisms serve as structural overlays connecting the relational architectures, for example to integrate IT and business processes [11]. Their primary role is to coordinate the various relational architectures [10]. As a coordination mechanism, they serve to highlight the complementary role that IT capabilities play in achieving business objectives and sharing common performance objectives [10]. Integration architectures also help in managing stakeholders’ perceptions of the IS function [29] and establishing its intra-organizational boundary [10]. In this way, they can also help provide a sense of identity to the IS function [10].

The elements and concepts of the platform organization are adapted in this paper to form a framework that theoretically analyzes how the company in the case study was able to successfully sense changes in its market and nimbly respond with new product and service designs. These mechanisms allowed the company to manage changes in processes involving products designed for different business units and systems designed for multiple business processes. While some of these organizing mechanisms may fall into the categories discussed earlier, such as overlays, we argue that the collective use of multiple organizing mechanisms was the key to the company’s success. Each organizing mechanism promotes a specific capability, which simultaneously enabled agility and innovation. Based on the modified platform logic [10, 11], the framework explains each capability, the way in which its delivery was governed and organized, and the way in which the arrangements facilitated working with and aligning delivery of IT with business objectives.

3 The case study

3.1 Methods of data collection and theory development

Since the goal of the study was to gain greater insights into an area not well understood in practice, and not well

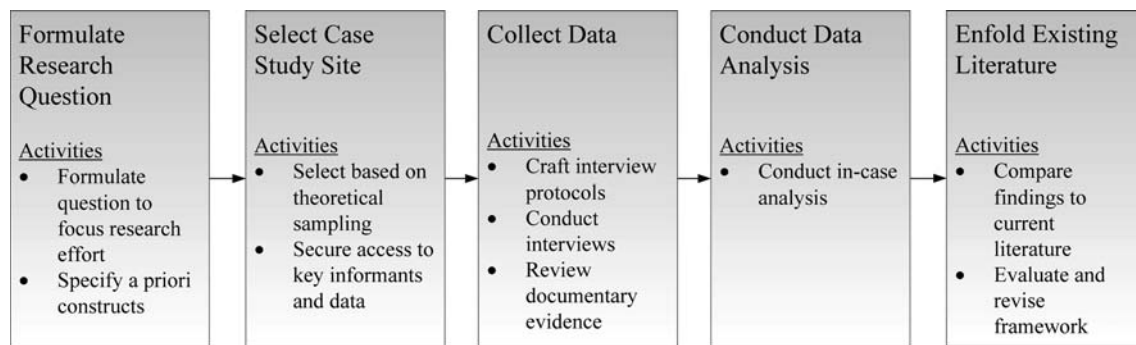


Fig. 2 Framework for case study theory development (based on [31])

explained by theory, we selected a case study to explore this phenomenon and to develop our understanding of the way one organization developed and managed agility [30]. Based on a prior call for research on platform organizations [10], the intent of this research is to populate the theoretical framework presented and thereby contribute to the development of this evolving area of research. The study followed Eisenhardt's [31] framework for developing theory with case studies, which is presented in Fig. 2 above. We began the process by formulating a research question to focus our research efforts [31]. Because the organizational motivation for change relates to the design of products and services, and the objective of our analysis is to understand the evolution of an IS management process to promote agility, the question posed is: how can changes in products and services, embedded in the information systems of a multi-unit organization, be managed in a way that promotes both global and local agility, allowing effective and efficient adaptation at both the global and individual business unit levels?

The company selected for the case study site was chosen for several reasons. First, the company relies heavily on information as a critical component of its products and services. Second, management had decided that agility was important to the company's future success. This offered us the opportunity to examine an organizational context where change is considered a critical factor for performance. By selecting a case study site that demonstrates these characteristics, we were able to adhere to the concept of theory driven sampling [31].

Access was secured through contacts made by the first author. At the time of the study, the company was experiencing large-scale system and organizational changes. To develop the relationships in the framework, information about change events was collected. This data allowed the researchers to understand the forces enabling or disabling changes to the product development and system implementation processes.

Multiple methods of data collection were employed, and included 23 unstructured interviews (10 in 1997, 13 in 2001,

with follow-ups in 2003 and 2004), internal documentation and reports, published articles, and historical reports about IS development efforts written by a senior employee. All interviews were conducted on site at the company, lasted one to two hours, and were recorded and transcribed. Interviewees included product managers, customer service operations managers, IS executives, IS development managers, IS project managers, programmers and testers.

Notes were taken during interviews, with more detailed accounts documented immediately afterwards. Follow-up interviews were used in several instances to confirm or expand on the evidence collected. A senior manager at the site reviewed the initial case write-ups to ensure the accuracy of events.

The in-case analysis was conducted in accordance with Eisenhardt's [31] guidelines. This involved writing up the case and condensing the extensive amount of data. Analysis continued throughout this process. Evaluation of consistent and conflicting findings in the literature lead to more detailed development of the platform logic framework and eventually to the final version presented here.

The analysis using the platform logic framework was developed by examining the interview transcripts to identify the various efforts described by the interviewees as having a positive effect on the company's efforts. While a great deal of the credit was attributed to reengineering information system applications and standardizing product design, interviewees also attributed success to a number of changes in the organizing arrangements. Further analysis led to the identification of these arrangements, resulting in a review of the literature to evaluate how the changes in arrangements could be explained and to inform current theory and practice.

3.2 Company and Industry Background

Delta Insurance¹ was founded as a small auto insurer in the Midwest, marketing its products through a network of

¹ The name Delta Insurance was created to conceal the actual name at the request of the company.

independent agents. The company initially targeted the non-standard or high-risk auto insurance market, which was often ignored by many insurance carriers. This segment of the auto insurance market became very competitive, however, as increasing numbers of non-standard carriers began to emerge over time. Because insurance is regulated at the state level in the United States, each state effectively constitutes a different geographic market, with different regional and national competitors selling through independent insurance agents in each market.

Price plays an important role in the competition between carriers, as it offers the most direct way for customers and sales agents to compare policies. This approach, however, requires that the many and varied policy components of different carriers be similar enough to enable price comparisons. This is often complicated by the numerous components that influence policy price, which include financial limits offered by a carrier for bodily injury, the availability of collision and uninsured motorist coverage, discounts offered to customers, percentage of payment required as a down payment to activate the policy, and the number and schedule of payments over the life of the policy. In addition, components that influence price (which are most often only noticeable to the other carriers or to savvy sales agents) are the underwriting rules for accepting a risk and the rate algorithm that sets the price. Together, these components constitute the complex design of an auto insurance product.

Typically, insurance carriers use underwriting rules as a design feature to control the mix of different risks in their overall risk-pool. Since non-standard carriers are often willing to sell to any-and-all risks found in any market, they often use pricing instead of underwriting rules in order to draw and detract different risks. By raising or lowering the price charged to insure certain risks, non-standard carriers in effect put in place similar controls, thereby influencing the mix of customers with various levels of risk in their overall portfolio. The market, of course, will also influence the composition of a carrier's risk-pool, reducing some of their control over risk. For example, comparative prices will influence the allocation of risks across all carriers. As a result, shifts in one carrier's pricing strategy may leave a competitor in an unwanted position, selling more policies than prudent in undesirable geographic territories and to higher-risk customers. Responding to these market conditions is often critical to a company's sales and profitability.

Further complicating matters, differences in state regulations and case law create tremendous variations in rules that govern underwriting, pricing and the collection and use of information. For example, minimum limits for insurance vary widely, as do the ability to use a customer's credit history as a factor used to evaluate a risk and price a policy. These and other variations in state statutes and case law create design challenges, but also opportunities, allowing

competitors to target creative solutions to certain market segments.

The highly situated nature of state regulations and the local mix of regional and national competitors, means that maintaining comparable product designs and accurate pricing in each market is difficult and potentially volatile. As a result, the ability to make changes to the insurance product's design is, at times, critical to a carrier. When an opportunity is identified in the market, the carrier must institute a product design change quickly, or incur an opportunity cost and potentially a loss of additional revenue. On the other hand, continuing to sell policies found to be unprofitable based on the latest policy claims data and calculation of present and future loss ratios, can negatively impact future profitability. As an insurance company's understanding of the risks they are insuring evolves and as sales influence the carrier's overall risk-pool, the desire to alter and refine a product design will grow.

3.3 Company strategy and structure

Delta's product strategy was to 'offer a price for every risk' and to 'wage a war by winning one state at a time.' The first part of the strategy meant that Delta competed primarily on price, designed within and implemented through elaborate product designs. Product design in this organization meant the collection, organization, use and application of data used in assessing and pricing insurance risks. These product designs often employed innovative use of customer and market data.

In offering a price for every risk, Delta assumed that every risk had an associated premium at which it could be profitably insured. This approach differed from competitors, who tried to eliminate high-risk customers up front and then use simpler approaches to pricing risks. Delta's approach had an advantage in that it guaranteed that the independent sales agents who sold their product were assured that they could always write their client's policy with Delta. The only question was how competitive Delta's price was relative to other carriers. With this alternative approach, accurate pricing was vital for Delta to control its overall risk-pool and remain profitable. The challenge then became identifying, collecting and analyzing the data necessary to accurately price each risk. Such a challenge required detailed knowledge of the local markets.

The second part of their strategy was to understand the local market risks and then design products in ways that were innovative, competitive and profitable. To execute this part of the strategy, Delta had historically organized around a product manager structure, assigning a manager to each state and allowing them a great deal of autonomy with respect to product design, pricing and marketing. Their approach differed from those traditionally used by standard

carriers, who organize around various functional specializations, such as underwriting, actuary, marketing and sales. The product manager model pooled these functions under one manager, resulting in a highly decentralized organization with many individual business units, which were well suited to understanding the local markets and state regulations.

Given the digitized nature of insurance products, IT played a critical role in all aspects of Delta's product design and delivery strategies. Delta had adopted a centralized IT model, with IT infrastructure, networking, application development and other IT-related functions residing within this centralized IT department. Delta structured their centralized IT organization around the various applications used to sell and process policies, such as policy sales, billing and claims systems. Twelve or more applications were impacted by a design change to a single product. Thus, the coordination of the design changes to the various systems during a single project for a business unit was a key challenge in managing changes in product design.

While Delta centralized IT to achieve efficiencies through economies of scale, the decentralized product management structure existed in order to provide greater flexibility and responsiveness to local market needs. Since management of the product design was highly decentralized, skills and expertise regarding information systems designs evolved along state and regional knowledge specific to the product design, even though programmers and testers were organized by the applications they worked on. This situation existed for most of the applications that supported policy sales and maintenance. How to manage the coordination of product designs across the different business units in order to establish standards and deliver new designs to market en masse was a key organizational concern addressed in the following sections.

3.4 Evolution of IT capabilities

By the early 1990's, Delta's auto insurance product was sold in over 40,000 independent insurance agencies, more than any other carrier in the U.S. The company was also very profitable and had attained a position among the ten largest carriers of auto insurers, based upon written premiums. In an industry that was historically a leader in IT usage, Delta excelled among its peers as an innovative user of information and technology. The company was consistently listed in general business periodicals and in its own industry trade publications as an IT leader.

The capabilities developed by the company during this time, with regard to product and information system design, were largely focused on the company's ability to devise solutions that were market-specific. This limited their ability to leverage solutions across multiple business

units, since state product designs were increasingly heterogeneous. Several factors explained the differences across product designs, as outlined by a senior project manager:

State regulations was the big one. [If] state regulations said we had to do something, we had to do that. Another was our competitive environment. If there was a competitive need we had in a particular state we would do that, we would design to that... if there are certain things competitors are doing at the state level that we don't do, we kind of opt ourselves out of the game, so we respond to that. And experimentation. Experimentation is things that a product manager may have noticed in an analysis that in his state they absolutely believe would have an impact on the bottom line. And so they'll say, try it.

While the approach of designing products to meet local market conditions was credited by executives at Delta with helping to drive company growth and popularity among independent agents, it also created complexity in product design, as well as the information systems in which these products were embedded. From a design standpoint, product decisions were controlled by state product managers, who recognized that there were inefficiencies in not sharing their innovative designs with each other, as explained by an IS manager:

Every product manager had his own version of the product and did not share [ideas with other product managers]...We were all kind of inventing wheels ourselves. You know, doing the work ourselves. So there were not a lot of best practices being shared.

The design of the product influenced the design and development of the information systems that supported underwriting and pricing. In fact, for years, the product managers themselves were given control over the process by which product-related information systems development projects were managed, as explained by this IS manager:

[When a] product manager decided he wants to do something in a state—change rates or add additional functionality, [they] called up a programmer that he happened to know and at that level the programmer would get a tester and it would be a very small group, three or four people, and the product manager would be the director, he would be the person running that project and get it implemented—whatever he wanted, didn't matter, his choice. He could take as long as he wanted.

The decentralized control of product design also created difficulties designing and maintaining the supporting processing systems. In effect, this unchecked control by the

project managers created obstacles for standardizing products, developing best practices and capitalizing on shared work processes and procedures. The disparity of designs also created obstacles to the company's ability to efficiently manage its IS operations and to implement design changes to its systems. Overall, the lack of standardized products and processes added costs and complexity to the operation and maintenance of the systems, as this project manager describes:

The thing is the complexity, so the more changes [to the product design over time], the more things you do [differently, together] just adds complexity to the system. How do you document it, how do you remember it, how do you know [what's] out there, how do you [manage it]...you can get a lot of state specific type change [in the systems over time].

Furthermore, the variations in product designs began to inhibit the company's ability to respond quickly to certain opportunities and threats in the market place.

3.5 The desire to change globally

With the evolution of the non-standard insurance market, Delta faced increased competition. The executive in charge of product development explained:

As a company, historically we've had [several] pretty strong advantages. One is what we would call pricing segmentation or product segmentation. And we lost that advantage to some degree as competitors began to copy [our product designs] aggressively ...You can copy a [design] pretty quickly. It's not hard.

The competition's ability to copy a product design is in part due to the regulations imposed by most states that require a carrier to file their data collection and rate factors with a state authority, making them public knowledge. To counter the competition's efforts to copy their product designs, Delta's devised a new strategy built around the capability to quickly introduce product design changes at the organizational level, as explained by an executive in charge of product development:

If people can copy us, [then] the next best thing is to be able to change quicker than they can on a national basis, and pretty much differentiate yourself in terms of your evolutionary speed. The faster we can change the underlying program then the more nimble we can be in the marketplace.

However, executing this strategy required not only a change in the manner in which product decisions were made, but also a fundamental shift in the underlying IT

infrastructure and applications that were required to enable agility. As one executive explained:

[Information systems] can provide us with either an advantage, in terms of how fast we can deliver a product, or as a constraint if our organization is too slow to react or if IT just provides a bottleneck to change.

To implement their new strategy, the company needed to improve their capabilities regarding product design standardization and implementation. Delta set out to implement a standardized product design across all states, and thereby develop the capacity to implement global change faster. The company's management knew standardizing the system would not be easy. While agreement on most of the design was possible, they would still need to accommodate the differences in the local regulatory and competitive environments. They also needed to be careful not to deter innovations, which were a source of competitive advantage that often originated from product managers as solutions specifically designed for an individual units.

The implementation process was complicated by familiarity with existing solutions and an expected resistance to change. Product managers often lobbied for and were frequently granted exceptions to the new standards, thereby undermining the company's future ability to change product design en masse at an organizational level. Managing the tendency to customize the standard product design was ultimately addressed in the way that the implementation efforts were coordinated and managed. Revising its approach over time, the company altered its organizational structures in several ways that ultimately facilitated the implementation of product design standards and their continued evolution. The success of the new approach, in the different phases, is described by a project manager:

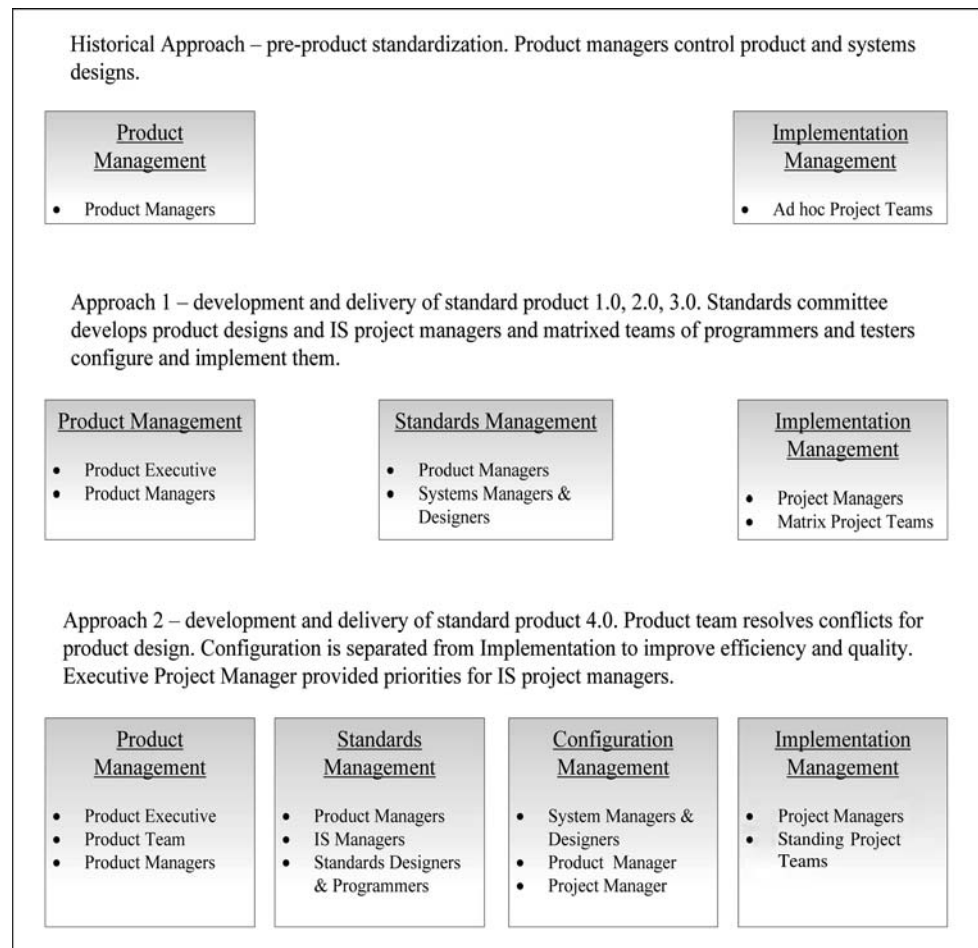
We worked on [implementing the 1st standard product design] for two years and something. I'm not sure we ever really finished it all. [For the third version], I think we finished most of it [in] sixteen to eighteen months ...[the fourth] we rolled out in eight months.

Speeding the delivery of the standardized product designs to market gave Delta the agile capabilities they desired, as explained by this IS manager:

Just to be able to quickly respond to the market and market needs, to our profitability needs. The advantage is if you can get there with your changes faster than the guy behind you can catch up, you [win].

The evolution of Delta's approach to designing and implementing these products is summarized in Fig. 3. Delta ultimately adopted the arrangements shown at the bottom of Fig. 3. The new standard design for products

Fig. 3 Evolution of product and system development at Delta: Historical, Approach 1 and Approach 2. Each change in process altered their approach to the integration and governance of the product design and their implementation in the information systems



helped the company achieve its vision of agility, but speeding the time to market for global design changes was also attributed to changes in the organizational arrangements used to manage the product's design and implementation. The top portion of Fig. 3 shows the historical approach, which simply required the product managers to work with IT to develop and deliver product individual designs.

The next approach is just the first used at Delta to implement its first three versions of the standard product design (versions 1.0, 2.0 and 3.0). Unfortunately, this approach resulted again in a great deal of customization to the design for each state's product, which slowed the process and ultimately resulted in yet another set of heterogeneous and complex applications. Eventually, Delta revised its organizational approach in a way that better facilitated the implementation process. This final approach, employed while implementing version 4.0, proved very effective in developing, adapting and delivering the new standard design. While the successful creation of a product design standard is credited with facilitating faster execution of delivery, the new arrangements put in place to design

and deliver the product also contributed to Delta's new capabilities for delivering solutions both globally and locally across its organization.

3.6 Achieving agility through governance

While the implementations of the second and third product versions were still underway, Delta began to plan for the introduction of product design 4.0. The delays in the implementation of earlier versions were very frustrating. Many of these projects took well in excess of the 12 week timeline originally targeted. Some progress was made between version 2.0 and version 3.0, with the new version 3.0 implementations taking a little less time, but the product changes between 2.0 and 3.0 were not as significant as those expected in the next version. Another long and protracted roll-out to all 48 business units was expected for version 4.0. Delta's analysis concluded that the processes for making design decisions and effectively coordinating design efforts between the business and information systems functions needed to be addressed. As a result, the following changes in structure and process were made.

3.6.1 Product management

To address issues associated with standard product design and timely decision making, a committee composed of three seasoned product managers and several senior business analysts was created. Reporting to the executive responsible for Product Management, this arrangement provided more effective direction for the new product design, as discussed by a product manager:

Probably the biggest thing that happened with the product, we had a more centralized group of people at the product level, [that is] the business level that controlled what was going to happen.

For the IS department, this team facilitated and clarified design questions, which helped to expedite their efforts in developing the new products, as described by an IS manager:

They offered solutions, offered advice from the business side, and kind of dictated things that would happen. So from that point on we didn't go to [individual] product manager and say what do you want, we went to the product team and said, this is what we've got and this is what [product managers say they] want more of, what should we build? There was [now] a different level of authority.

This arrangement also provided greater control over the evolution of product and system designs. While the desire to innovate and experiment based on the environment of the specific business unit remained, the centralized product team became the governing body for research and development, serving as a control point to aggregate new ideas. They would also approve or deny customization of unit-specific product designs. As a member of the product team describes:

They put a...product team in place that said [to state product managers] if you can't give us a valid state-required reason for making this difference, then, no, we will not allocate you [the IT] resources [to do it]. [Now] we could kind of control [product design standards].

This control was applied in a way that provided the ability to continue the development of product design innovations, while curbing the tendency for design variations to constrain Delta's ability to introduce new designs globally across the organization or share design innovations across the business units. This approach is described by an IT manager:

We're trying to do the best of both worlds at this point. We've got this centralized commitment,

[which can say] sorry, no variations because there shouldn't be [any ideally when the] product is centrally developed. The [product managers] in the field have the ability to give input on how they want to tweak their product and experiment with it. The central product team will say, OK, that [customization of the product design] sounds good, if you have a basis for it, then it sounds reasonable, then let's try it.

And if it's good, if it's really good, it probably works for other people, so it will get into the base product. We still have a standard approach but we're trying now to accommodate the variety that comes with the different regulators and geographies that we play in.

As a product manager described, the process increasingly made them think first about what they might propose, and ultimately helped to bring out better ideas:

In my mind, anyway, [this] tension is for the most part probably healthy because, if you know there's going to be scrutiny, you are more inclined to do your homework, I think. And therefore somebody won't come running in with a cockamamie product idea. They'll think long and hard about it. And the product team isn't going to just squelch it because they didn't think of it, either. So I think there is a balance and there is some negotiation that takes place. But I think it works.

3.6.2 Product standards management

While the composition of the committee managing the design and development of product standards remained the same, changes in the way the committee members allocated resources to the implementation projects had a significantly positive effect on maintaining standards. Previously, when the product design could not account for all of the business unit's specific needs, programmers would need to customize and build a solution. Alternatively, a product manager might have reasons to want to try out a new approach in their product designs, as described below by an IS manager:

Let's say a state needs some kind of discount, because its state mandated. So normally we'll try to figure out which [other] states need this. If five states need this we try to build a structure that builds this five different ways, so my five states can [use it]. But if state number six comes in and says, I want that, it's not required, but it would be good in my state, ... [but] I think I'd like to do it [a different way]...we're

like, no, no, pick one of these [five]. Yet, if you could come up with the numbers to prove that this was a good thing, we might do it, it's not like it was hard and fast. If you could come up with some valid analysis that said this would help [performance] and show the numbers [we'd do it].

To manage this process, the project manager would coordinate with representatives of the standards committee. Under the original approach the standards committee assigned liaisons to each project. The liaisons were charged with making recommendations for designing as standard a solution as possible, allowing for reuse in other business units. With programmers performing the actual design and implementation functions, problems occurred when different approaches to similar problems were designed by different liaisons, duplicating efforts. This was due in part to the lack of coordination and communication between the liaisons, but more so it was due to each liaison undertaking many areas of specialization involved in a unit's product design. Since each liaison would design customized solutions to many product design features, the opportunities for differences in approach across product designs grew.

A new approach was devised, this time using ad hoc teams of liaisons from the standards development committee. Each liaison had an area of product design expertise in which they would design and develop solutions. By creating ad hoc sub-teams composed of specialists, who would focus their individual efforts on their areas of specialization, the approaches to customizing design across business units could be more standardized. This not only promoted greater uniformity of design across units, but also increased the chance of reuse and permanent incorporation of new designs into the standard product and system designs.

3.6.3 Configuration management

Individual projects were required to configure and implement the new product designs into each business unit. In the initial phases, these efforts were conducted by large project teams that reported within a matrix structure to an IS project manager. Each project required the formation of a team responsible for gathering requirements and configuring the system, then programming, testing and implementing it.

Management believed this process could be improved by making changes in how responsibilities were assigned and the teams were structured. As a result, configuring and implementing the product and system designs were approached as different steps in the process. Furthermore, additional layers of authority and decision making were

added. Third, certain processes were modified to insure that information and decisions were effectively communicated.

The changes were instituted when managers determined that stabilization of specifications was a necessary precursor to faster implementation of projects. As an IS manager described:

Our programming is actually a week to two weeks for a whole revision. The majority of it is in the definition stage, getting the product manager to nail down his requirement...

To accelerate the process, a new arrangement was developed to help each state define their product and service designs. Rather than involving all of the programmers and testers in the process, a small group would meet composed of the product manager, his marketing assistant, the IS executive in charge of the reengineering, the IS project manager, and several business analysts assigned to the product team. Instead of the project manager, the IS manager in charge of the overall reengineering process controlled each of these meetings. The product manager from each business unit and their team would work through the checklist themselves to specify the requirements, while the business analysts and the project manager guided and instructed them through the process.

This new approach was credited with the creation of much more complete and specific state requirement definitions, while minimizing meetings and follow-up. This focused effort resulted in not only shortening the overall elapsed time for each project, but also providing the project manager and their staff with a much better grasp of product design and implementation intricacies.

3.6.4 Implementation management

Previous efforts had also demonstrated that coordinating the timing and delivery of projects across 48 different business units in a short period of time was a daunting proposition. Management realized the need to control the priorities for assigning the resources and scheduling of projects. Managing project scheduling and prioritization was now governed by the product executive responsible for the product managers, through a liaison.

Rather than having IS project managers negotiate with product managers for resources as before, the product executive now had the authority to coordinate priorities across the state product managers. This was needed to ensure that when the time came, each was prepared to move forward with their state's implementation. Each state was given a slot in the master schedule, which was likened to a bus schedule. One IS executive described how product managers were told:

[A product manager was told they] have a time slot when they are scheduled to be on a bus, which is scheduled on a specific date and time. If you can't be ready for that bus, you will have to wait for a later bus, but we can't hold the bus for you.

The executive, in discussing the bus metaphor, stated that a product manager could take a later slot in the master project schedule if they were not ready. The current schedule could not wait for them, so another business unit could then use that time slot. As the product executive acted as a single point of contact for project coordination, this additional layer of authority was credited with reducing conflict and shortening the overall elapsed time for project implementation.

Lastly, to promote greater continuity in the project teams, Delta discarded its matrix approach to managing project resources and created permanent cross-functional co-located implementation teams. These teams worked together on an assigned set of implementation projects. During these projects, each team was co-located. Having permanent teams reportedly built strong working relationships across the team's members, as described by an IS manager:

We had a compressed [time line] and ...cycle. And what we had done is set up a kind of assembly line approach to changing [product designs] around the country. ...we had dedicated teams that were going to go through a series of similar [projects], so that there would be [a reduced] learning curve, so that the first [business unit] would probably take a little longer than the second, the third, or the fourth. They get pretty good at what they were doing.

While each project was still managed by an IS project manager, co-locating the teams allowed the design process to be more responsive to problem solving and design changes. This approach reportedly yielded substantial benefits as product managers needed to consider various alternatives to solving a product or process design problem. A project manager could easily arrange and conduct impromptu meetings, whereas with the earlier matrix organization such meetings required the project manager to traverse a host of scheduling issues in order to assemble a team.

4 Data analysis

In the following analysis, we examine the evolution of the product and service design and delivery processes in the case study. Our analysis outlines the distinct capabilities the organization historically leveraged and those developed

during the time outlined in the study. We outline the design shortcomings of the organization's initial arrangements to provide a context for analyzing the evolution of delivery arrangements as the case study unfolded. We then examine how Delta successfully structured and governed the delivery of these capabilities using the platform logic [10].

We specifically examine how changes in Delta's organizational arrangements or organizing logic increase its sense and response capabilities and thereby improve overall enterprise agility. By using the platform logic, we subscribe to Sambamurthy and Zmud's [10] definition of organizing logic as "the managerial rationale for designing and evolving specific organizational arrangements in response to an enterprise's environmental and strategic imperatives" (p.107). In the case of Delta, the imperative is to improve enterprise agility, but more specifically, Delta wanted to increase its IT capability to deliver organizational-level responses in product design en masse across all of its business units.

In applying the platform logic, we identify relational and integration mechanisms. As defined earlier, relational mechanisms represent arrangements that enable an organization to effectively source, configure and deliver IT capabilities. Alternatively, integration mechanisms represent arrangements that enable the delivery of capabilities by enabling coordination of activities across organizational functions, especially those that link IT and business functions.

The analysis will unfold by outlining each of four relational mechanisms (i.e., product management, standards management, configuration management, and implementation management). For each mechanism, we outline the capabilities desired and the historical arrangements formed to support them. We also explain the shortcomings of the historical arrangement. The analysis proceeds in describing how Delta appropriated and adapted these arrangements over time to improve its response capabilities.

4.1 Product management

Historically, product management played a critical role in the design of the information systems discussed in the case study. We view product management as an intra-organizational relational mechanism, which establishes important relationships between the business and IT necessary to source, configure and deliver IT capabilities [10, 11]. Whereas many organizations utilize IT departmental liaisons to elicit requirements from business managers, Delta encouraged its product managers to engage directly with IT. As a result, product managers learned to direct the design and manage the development of Delta's information systems.

Initially, product management was controlled independently within each business unit, giving product managers responsibility for sensing environmental change and initiating responses, often in the form of changes in product design. In this way, product managers initiated and even managed many information system projects. However, the decentralized governance of the business units created tremendous variations in product designs across those units. The resulting complexity in the systems slowed Delta's response capabilities when implementation of a standardized product design became an imperative.

Delta's approach to product management enabled two important IT capabilities. First it provided *knowledge work leverage*, that is "the sharing of knowledge and the joint ownership of technology based initiatives" [10] (p.109). As the critical resource best suited to inform the design of the systems that supported a business unit's products and services, product managers provided IT with knowledge, direction and insights for innovation.

Greater knowledge of information systems design also afforded Delta's product managers the ability to create *value innovation*. Value innovation is "envisioning how novel combinations of business levers could be realized through IT and thus promote IT-based business innovations" [10] (p.109). Product managers created numerous innovations in the use of information and technology at Delta. However, design variations across units complicated efforts to share innovations across those business units or implement a standardized product design.

Applying the platform logic, we view product management as an important relational mechanism that provides *knowledge work leverage* and *value innovation* capabilities. Initially, individual product managers provided the business knowledge and direction for IT to design systems. Resolving problems related to the coordination of product and system designs across business units, however, required additional integration mechanisms.

We view several changes made to the structure and governance of product management as important integration mechanisms, which enhanced Delta's ability to coordinate product design and delivery with IT. The product executive and product team both helped to coordinate work across business units in ways that enabled IT to develop and deliver new product designs.

The product executive's role was to provide a vision and leadership for unifying the disparate approaches to product designs. The product executive also provided leadership for the implementation effort, by establishing scheduling priorities in order to keep the overall standardized product introduction on schedule (see discussion of Implementation Management below for more detail regarding implementation problems).

Governing the design and implementation of a standardized product proved difficult for a single manager, especially with respect to the timely processing and resolution of appeal decisions regarding enforcement of standards. To speed decision processes, Delta created a product team composed of three former product managers. This arrangement enhanced their ability to respond quickly with new product designs at the organizational level.

As shown in Fig. 4, product management is a key relational mechanism. This relational mechanism is enabled by two integration mechanisms: the product executive and the product team. Together, these mechanisms worked to govern product management and coordinate design standards across the business units.

4.2 Standards management

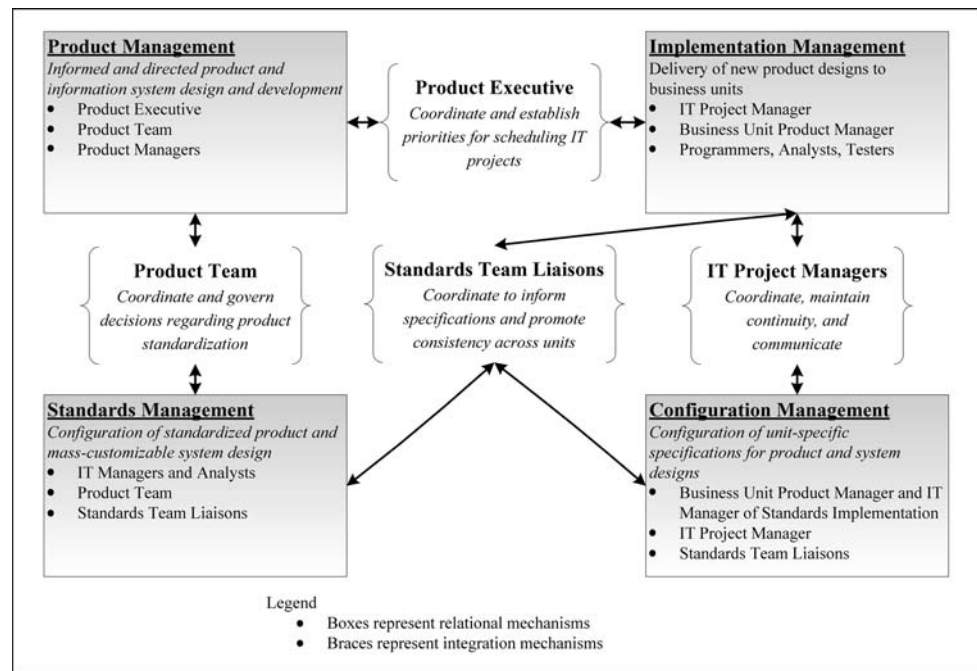
Responding to environmental change at the organizational level necessitated a new capability: *providing an IT-enabled business platform* capable of introducing changes to product designs en masse. We view standards management as an important relational mechanism that enabled this capability and promoted greater enterprise agility.

Standards management was initially managed by a team of experienced IT managers and business analysts, many with highly specialized knowledge of the relevant IT application. In spite of the experience and expertise of this team, two persistent challenges existed. The first was that a mass-customizable product design had to support a wide range of requirements to meet the needs of the different business units. Second, significant challenges existed in trying to collect and confirm all of the requirements necessary to satisfy all business units and prevent excessive customization of the standardized product design during the implementation process.

The product team fulfilled a key role in the decision making process. The experience and expertise of these former product managers, coupled with their authority in governing product management gave them the ability to expedite decisions. In a process where problems arise continuously, and their timely resolution is critical for success, the product team's coordination of standards with product management was a valuable integration mechanism.

The team also needed to coordinate standards with the implementation projects, as the need for customization occurred. Initially, the standards team assigned a liaison to each individual implementation project. The liaison could approve and design all customizations for a single project. Problems arose, however, as these designs varied across projects. Delta resolved this issue by assigning liaisons based on their expertise, rather than asking them to be generalists and address all the design questions related to

Fig. 4 Integration and relational mechanisms realized in Approach 2



an entire project. This arrangement was instituted for the two applications where design changes to products were most prevalent.

The liaisons represent an important integration mechanism, coordinating the design and implementation of the standardized product design with individual business unit. This arrangement allowed Delta to manage the standards more effectively. By representing the standards team in the implementation projects, the liaisons exercise governance over the customization of the standardized product designs. Furthermore, by vetting and managing design decisions, Delta devised clearer boundaries regarding decision rights and authorities.

4.3 Configuration management

The implementation of the standardized product into each business unit required configuring each instance of the product design to address the needs of a particular market. We view the arrangements employed in conducting this process as a relational mechanism for sourcing, configuring and delivering two critical IT capabilities: *leveraging knowledge work* and *delivering value innovation*. We examine how Delta altered the arrangements used to structure and govern this mechanism and improved the effectiveness of the configuration process.

Initially the configuration process began by asking the product manager to complete a long checklist of standardized design choices based on business unit requirements. The IT project manager would then review the checklist in the implementation kickoff meeting, where

the entire team of programmers, analysts and testers assigned to the implementation team would review and vet these decisions. As described in the case, this process was deemed a source of delay when design issues were not quickly resolved. IT managers believed that product managers were too unfamiliar with the new product designs, so they needed to be better educated about standards.

Changes to the arrangement in the configuration process included separating it from the implementation project and limiting the number of participants. The kickoff meetings were now led by a senior IT manager who had in-depth knowledge of the standardized product design. The senior IT manager assisted the product manager and their staff in completing the checklist. The smaller group was reportedly more effective at completing the checklist, thereby stabilizing the requirements and speeding the implementation process.

Two additional parties attended the product configuration meetings and helped to coordinate requirements with other parts of the organization. First were the IT project managers who led the implementation projects. They provided continuity and coordinated project efforts across these now separate and distinct groups. Also in attendance were two key business analysts, who worked closely with standards management. The analysts informed the configuration process through their knowledge of the standard product design and helped to coordinate configurations across the different business units.

The two business analysts and the IT project managers constituted integration mechanisms that helped Delta coordinate work across other organizational areas involved

in the delivery of the new product designs. As mechanisms for coordination, the analysts informed decisions related to standards, while the IT project manager helped to communicate and clarify requirements defined in the configuration process with the execution of the implementation process.

4.4 Implementation management

Implementation was a critical function necessary to support Delta's IT capabilities related to *solution delivery*. This capability is defined as "managing assets such that continual streams of IT-based solutions are provided in response to emerging business opportunities and challenges" [10] (p.109). Historically, implementations were sourced using dynamically configured teams of programmers, analysts and testers. IT managers heading various IT functional application areas allocated resources to these projects based on availability. In this matrix organizational structure, team members reported to both their respective departments as well as to multiple IT project managers, depending on the number of projects to which they had been assigned.

Persistent problems existed in staffing these teams. First, limited availability of a resource in one functional area could delay the execution of a project. Second, team members did not work together regularly, so there were limited opportunities to develop effectively as a team. Third, team members sometimes lacked expertise in the assigned business unit so a steeper learning curve existed. As a result, team performance was inconsistent.

Delta altered this arrangement by disbanding the use of dynamic teams and organizing standing teams led by a permanent IT project manager. These changes in the team structure enabled a single point of governance for resources, thus providing a relational mechanism more capable of sourcing and delivering a critical IT capability. The teams also worked together continuously on concurrent projects, giving them opportunities to share knowledge and develop a team identity. This change was credited with more efficiently delivering solutions.

As with the other relational mechanisms, integration mechanisms also contributed to the improvements realized. First, implementation management benefited from the integration mechanisms discussed in the standards management section above. Dynamic teams of specialists now represented the standards team on a project, rather than using individual liaisons assigned to each project. This new arrangement helped in coordinating design standards and customizations within and across business units.

The second integration mechanism, also discussed earlier and elaborate on it here, is related to the complexity of introducing a standardized product across all 48 business

units. Creating and resolving conflicts in a master schedule presented a difficult challenge for the IT department, since product managers previously controlled the schedule and implementation of IT projects related to product design themselves.

Quickly executing this many projects within a limited time span presented increased complexity, especially when business units experienced delays related to pricing decisions or regulatory approval, forcing projects into an on-hold status. Delta could not afford to have an implementation team standing by awaiting these decisions. To help resolve these issues, governance over scheduling was granted to the product executive. His office coordinated with IT in setting priorities and assigning projects to the six IT project managers. This integration mechanism improved coordination and helped to clarify the roles of product management and implementation management in the delivery process.

5 Discussion and conclusion

As companies develop strategies based on agility, they should expect that information systems positively affect execution of these strategies, and ultimately firm performance. Indeed, information systems can offer the capability to quickly redesign products and deliver them to market faster than ever before imagined. However, without mechanisms in place to control and focus the nature and scope of IT's evolution, complexity may be introduced into systems and processes that can stymie an organization's ability to quickly sense and respond to changes in its environment.

The purpose of the case study was to present Delta's success as a lesson in organizing structures and processes for developing and sustaining agility. In other words, Delta's true needs were to develop capabilities that enabled adaptation and evolution, rather than designing for system standards and a mass-customizable design. While system standards and mass-customizable designs were an important part of Delta's strategy and contributed to their success, these tools were not sustainable without developing the organizational mechanisms necessary to simultaneously address the demands of global organizational change, local business unit requirements and market caprices.

As with any research, this study has limitations which merit discussion. The main limitations of this research include those imposed by the method of investigation and the study's limited context. This in-depth longitudinal case study was conducted in a single organization which invites questions as to the generalizability of the findings, misjudging the representativeness of the events at the case site, and other investigator biases [32]. To guard against

these limitations, we endeavored to use theory-based sampling to choose the study site and multiple forms of data collection to triangulate our findings [30, 31].

The study's context is a multi-unit organization whose strategy centered on agility at both the business unit and organizational levels. Moreover, our organization marketed a digitized product that required standardization at the organizational level while remaining flexible and adaptive to local market needs. We contend, however, that the lessons learned from Delta's efforts to become an agile organization are applicable to other organizations that operate in a hyper-competitive, dynamic environment and are struggling with implementing a strategy based on increased agility and improved governance.

Despite these limitations, this longitudinal case study makes four contributions to the literature on agility and governance. First, this study extends the research on agility in multi-unit organizations. Our analysis provides insights into how agile organizations increase their ability to sense and respond to changes in their competitive environment. We demonstrate that achieving agility requires organizations to innovate, organize and integrate information technology and business objectives in often-times complex settings. As the case at Delta demonstrates, achieving agility is an iterative process that requires experimentation at multiple levels of the organization, requiring changes to products, services and the platforms that support both.

Second, this work extends our understanding of the platform logic of organizing and serves to answer Sambamurthy and Zmud's [10] call to research on complex organizations. As one of only a very few studies to have adopted this framework for studying organizing logics, it serves to populate and extend our understanding of this theoretical lens. Moreover, it reinforces Schwarz and Hirschheim's [11] observation that managing the relationship between IT and business is an important focus of governance research. Furthermore, ours is the only study we know of to apply this framework to examine and explain improvements in agility made in a specific organizational process. Using the platform logic, the experiences at Delta reinforce that organizations increasingly concentrate their efforts on managing relationships between business users and IT to deliver solutions while maintaining control over product and process evolution.

Third, this study enriches our understanding of integration mechanisms by expanding the role of liaison mechanisms in organizations, demonstrating how these can contribute to the creation of adaptive organizational structures. Delta's effective use of representatives from various business units and IT serves to illustrate how liaison mechanisms may be employed to create services

and solutions that are standardized at a global level while satisfying local level needs.

Finally, by demonstrating how governance can increase one organization's ability to sense changes in the environment and quickly respond, this research serves to strengthen the theoretical connection between governance mechanisms and an increasingly important organizational and IT capability—i.e., agility.

This research offers several practical implications for organizations grappling with the seemingly conflicting goals of developing global standards that promote efficiency while remaining flexible and responsive to business unit needs. This study provides practitioners with insights into how a large, multi-unit organization successfully executed a strategy steeped in agility by experimenting with new and appropriating existing mechanisms that enabled it to simultaneously adopt standards for improved efficiency, variation for increased responsiveness, and processes for timely delivery of products to market. In so doing, we provide guidance on how applications and technologies implemented today can continue to enable agility tomorrow.

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